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Evidence for a Neoarchean LIP in the Singhbhum craton, eastern India: Implications to Vaalbara supercontinent

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ABSTRACT

We report eight new Pb-Pb baddeleyite ages and paleomagnetic results on a series of hitherto unknown Neoarchean NNE-SSW trending mafic dyke swarms intruding the Paleoarchean basement rocks in the Singhbhum craton, eastern India. Being the most dominant dyke swarms in the region, they occur over the entire Singhbhum craton with an areal distribution extending to about 30,000 km^2. Of the eight dyke samples analysed across the entire dyke swarm, six were emplaced at 2762.4 ± 2.0 Ma (weighted mean of 6 dyke ages), two with similar strike directions, yielded older and younger ages of 2800.2 ± 0.7 Ma and 2752.0 ± 0.9 Ma respectively. The older 2800.2 ± 0.7 Ma dyke event appears to be coeval with several other magmatic events reported earlier from the Singhbhum craton. Emplacement of dacite tuffs within the Malaigiri basin, on the southern margin of the craton, the Budhapal and Rengali granite from the south-western and southern parts of the craton, along with the Temperkola granite activity and associated acid volcanics in the western part of the craton, and perhaps the Mayurbhanj granite (fine-grained phase) are also contemporaneous, suggesting a fairly wide spread thermal event in the region at this time.

Well defined craton-wide magmatic events contemporaneous to the 2762.4 ± 2.0 Ma Singhbhum dyke activity have been reported from the Pilbara. The younger 2752.0 ± 0.9 Ma age dyke event occurring in the Singhbhum is also know from the eastern Pilbara. An older event at 2783 ± 1.2 Ma (Gaborone-Kanye-Plantation Porphyry-Derdepoort-Modipe episode) was identified in the Kaapvaal craton. Comparison of paleomagnetic data obtained on the Singhbhum Neoarchean dyke swarms with the Pilbara and Kaapvaal data, show all these continents were located at steep latitudes and could have been proximal to each other during this time. If Pilbara and Kaapvaal formed supercraton Vaalbara during the Neoarchean time, we propose Singhbhum with similar Archean geology could also have had its ancestry in the same supercraton.

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1. Introduction

Precise geochronology on mafic dyke swarms in conjunction with paleomagnetism provides first-order information necessary for identifying large igneous provinces, for paleoreconstruction of crustal blocks and for revealing the history of supercontinent formation and break-up. As in many Archean blocks mafic dyke swarms are widespread in the Singhbhum craton too, and are known as 'newer dolerites' in the literature (Saha et al., 1973; Saha, 1994). Several K-Ar whole rock age determinations have been attempted on these dyke swarms, which vary appreciably within and between swarms and range from 2144 Ma to 950 Ma (Sarkar et al., 1969; Sarkar and Saha, 1977 and Mallik and Sarkar, 1994). A distinctly older Rb-Sr whole rock isochron age of 2613 ± 177 Ma (Roy et al., 2004) was reported for a NNE-SSW trending ultramafic dyke from the northern part of the craton. A Pb-Pb baddeleyite age of 1765.3 ± 1 Ma was recently reported for one of these dyke swarms with a NW trend (Shankar et al., 2014). Although these age determinations have shown that most of the "newer" dolerite swarms are Proterozoic in age and at least one of these could be Neoarchean, it is not clear whether the large spread in the measured ages from Mesoproterozoic to Neoarchean is real or due to inherent limitations of the dating methods (K-Ar and Rb-Sr) when applied to Proterozoic rocks.

U-Pb age determination to ~0.1% precision on mafic rocks has been possible mainly by the analysis of baddeleyite (ZrO_2) found as a common accessory mineral in them. Baddeleyite generally

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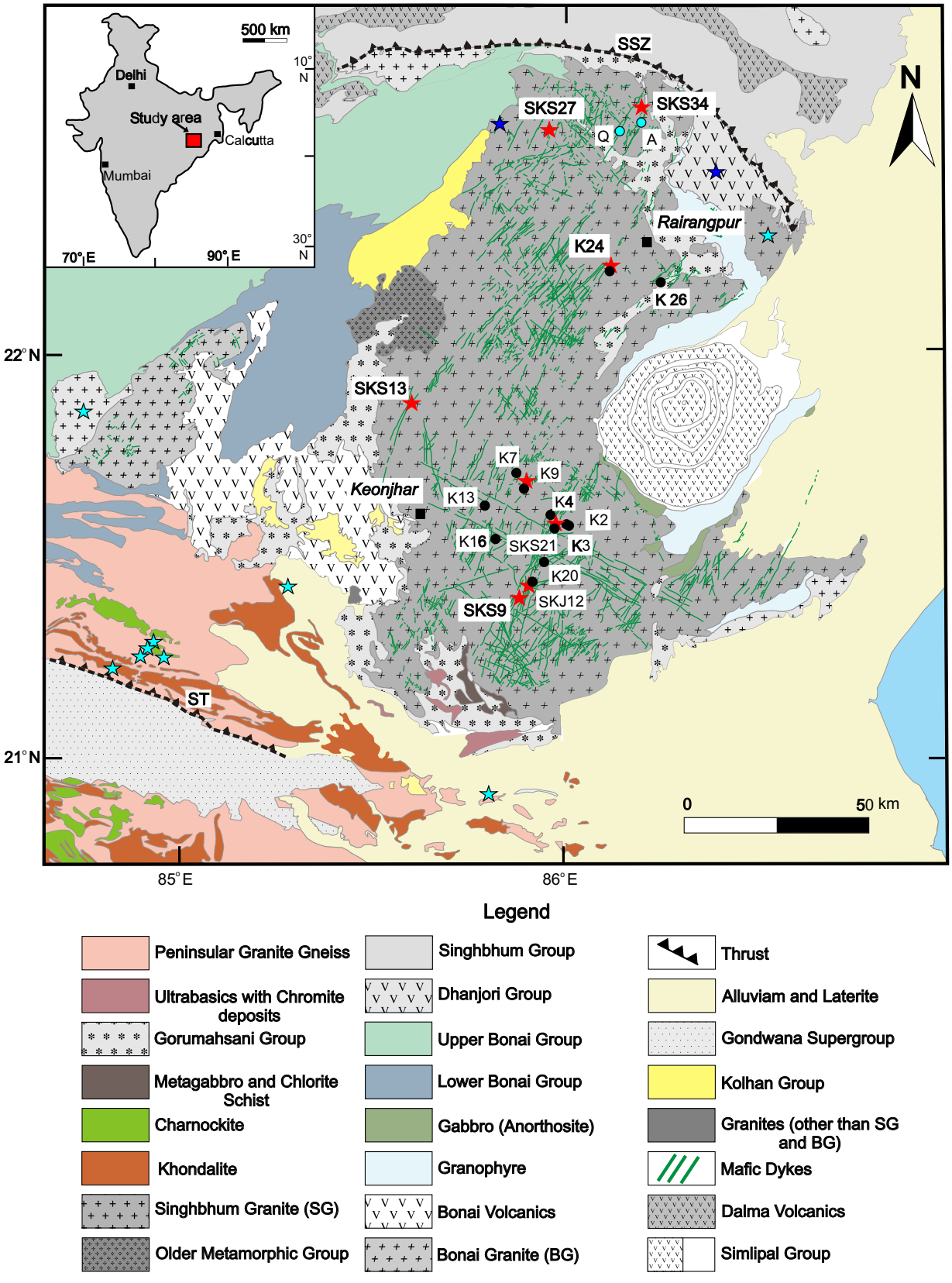


Fig. 1. Simplified geological map of Singhbhum craton (modified after geological quadrangle maps published by Geological survey of India: 73B(2005), 73C(2008), 73F(2008), 73G(1991), 73H(1985), 73 J(1998), 73 K(1998), 73L(1996)). Distribution of dykes traced after google earth satellite images. Black solid circles indicate the locations of paleomagnetic samples from present study, Cyan blue solid circles indicate the locations of paleomagnetic samples from Verma and Prasad (1974), red stars indicate the locations of dated dyke samples (present study), Cyan blue and Blue stars indicate the locations of previously published U-Pb/Pb-Pb (Zircon) (Nelson et al., 2014; Misra et al., 1999; Bandyopadhyay et al., 2001) and Rb-Sr whole rock (Roy et al., 2004) Pb-Pb whole rock (Misra and Johnson, 2005) geochronology sites respectively. SSZ: Singhbhum Shear Zone; ST: Sukinda Thrust. Inset shows the location of the study area in India. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

contains sufficiently high concentrations of uranium (500 to several 1000 ppm, Heaman and LeCheminant, 1993). It yields concordant or nearly concordant (less than 1% discordant typically) U-Pb data even without any of the pre-treatments typically required for zircon preparation, as it appears to be much less susceptible to Pb loss compared to zircon (Heaman and Tarney, 1989; Heaman and LeCheminant, 1993; Wingate, 1998; Li et al., 2005; Bayanova, 2006). And importantly it rarely occurs as xenocrysts in these rocks. Due to such exceptional qualities, this mineral is considered to be an ideal geochronometer for U-Pb or Pb-Pb dating of mafic and ultramafic rocks.

This paper presents eight new $^{207}Pb/^{206}Pb$ baddeleyite age determinations on the NNE-SSW trending 'newer dolerite' dyke swarm (Fig. 1) following the TE-TIMS (thermal extraction-thermal ionization mass spectrometry) approach (Kumar et al., 2014). In addition paleomagnetic studies have also been carried out on 12 of these dykes. These techniques are used here to determine their emplacement age, establish their extent in the craton, and determine the possible paleoposition of the Singhbhum craton in the supercraton configuration at that time, which could lead to infer its antecedent craton.

2. Geology and sampling

The triangular Singhbhum craton is bordered to the north by the North Singhbhum Mobile Belt (NSMB), the Eastern Ghats Belt to the southeast and the Bastar craton to the southwest (Saha, 1994). The major crustal units that make up the craton include greenschist- to amphibolites-facies supracrustals and tonalite–trondhjemite–granodiorites of the Older Metamorphic Group (OMG) and granitoids and tonalities of the Singhbhum Granite Complex (SGC) and the greenschist facies platformal sediments and banded iron formations interlayered with mafic and felsic volcanic rocks of the Iron Ore Group (IOG) (Fig. 1). Singhbhum granite complex includes the Singhbhum granite along with the Nilgiri, Bonai, Pala Lahara, Mayurbhanj and Kapdipara granites (Saha, 1994). Recent geochronological studies show that tonalite–trondhjemite–grano

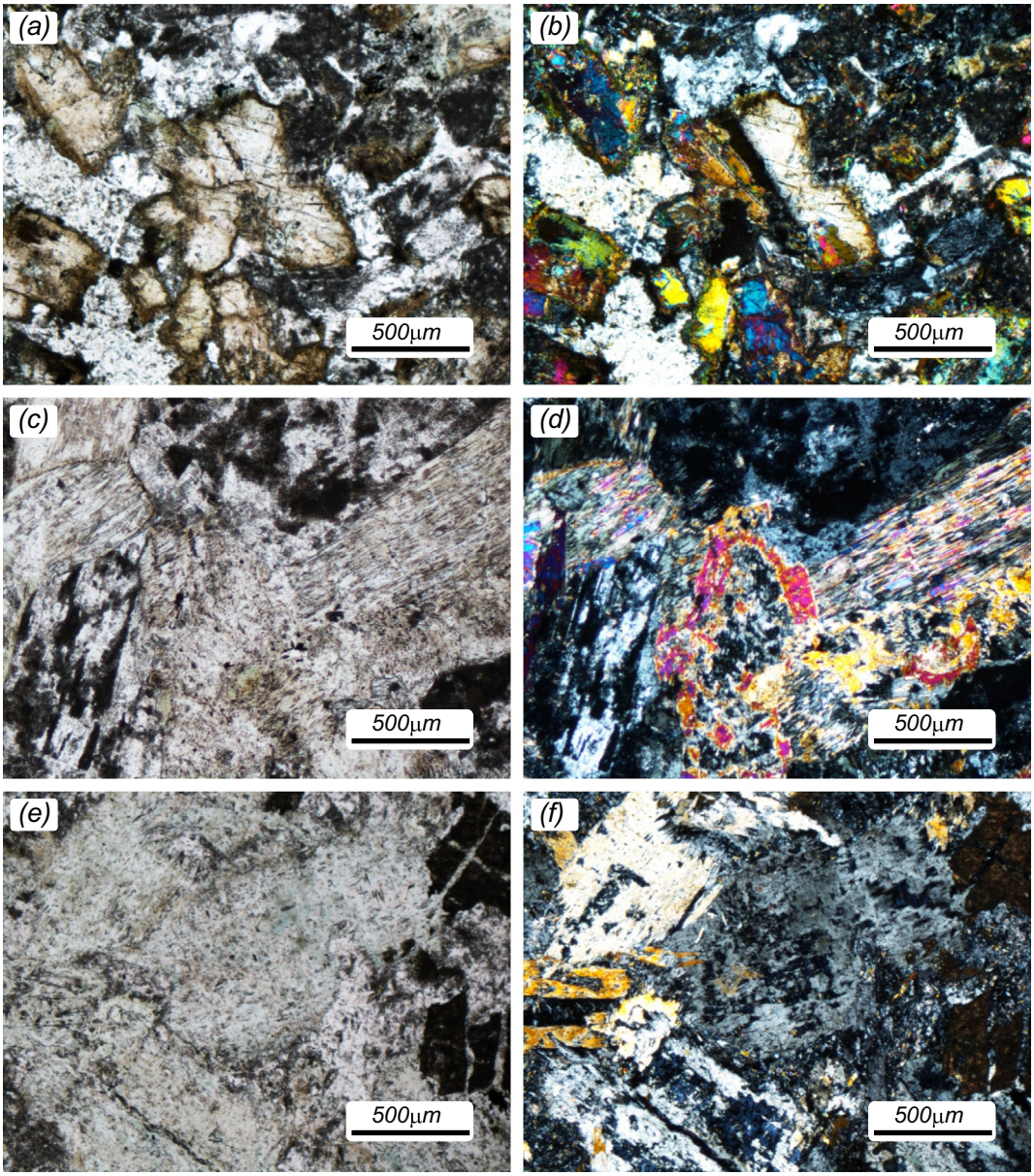


Fig. 2. Photomicrographs of NNE-SSW trending dyke samples. (a) and (b) are from site SKS 9, (c) and (d) are from site SKS 27 and (e) and (f) are from site SKS 34.

diorites of the OMG and granites of the SGC were emplaced episodically between 3.45–3.44 Ga and 3.35–3.32 Ga (Upadhyay et al., 2014).

Dykes swarms known as 'newer dolerites' occur within the Singhbhum granite complex. Petrologically these vary from doleritic, leucogranophyric to ultramafic to noritic in composition. Doleritic dykes are however the more dominant (99%) and outnumber the other types (Bose and Goles, 1970; Saha et al., 1973). The newer dolerites were emplaced mainly in three distinct strike patterns NNE-SSW, NW-SE and WNW-ESE. Based on cross cutting field relationship of these dykes, the NNE-SSW swarm appears to be the oldest dykes intruding the Singhbhum and Bonai granites

Table 1
TE-TIMS Pb isotopic data on baddeleyite fractions from the Neoarchean Singhbhum dykes.

Sample. No. (location)	No. bloc.	$^{206}Pb/^{204}Pb$ (m)	Abs. err	$^{207}Pb/^{206}Pb$ (m)	$^{207}Pb/^{206}Pb$ (c)	Age (Ma)
SKS 09 (21.389°N; 85.871°E)						
SKS 09-1	26	13639	277	0.193290 ± 070	0.192429 ± 065	2763.0 ± 2.1
SKS 09-2	20	17367	186	0.193164 ± 067	0.192468 ± 067	2763.4 ± 2.1
SKS 09-3	30	21497	261	0.193131 ± 048	0.192576 ± 047	2764.3 ± 1.9
SKS 09-4	18	20990	858	0.193188 ± 060	0.192601 ± 058	2764.5 ± 2.0
SKS 09-5	16	12949	122	0.193384 ± 063	0.192474 ± 058	2763.4 ± 2.0
SKS 09-6	28	18562	174	0.193131 ± 040	0.192478 ± 040	2763.5 ± 1.9
Weighted Mean age = 2763.7 ± 0.8 Ma, MSWD = 0.34						
SKJ 12-3 (21.401°N; 85.885°E)						
SKJ 12-3-1	30	6864	402	0.194421 ± 093	0.192489 ± 049	2763.6 ± 1.9
SKJ 12-3-2	29	9576	145	0.193954 ± 055	0.192684 ± 048	2765.2 ± 1.9
SKJ 12-3-3	26	25071	625	0.192993 ± 052	0.192505 ± 057	2763.7 ± 2.0
SKJ 12-3-4	16	24203	2955	0.193217 ± 093	0.192613 ± 108	2764.6 ± 2.4
SKJ 12-3-5	30	20928	633	0.193211 ± 067	0.192616 ± 062	2764.6 ± 2.1
SKJ 12-3-6	22	29748	1877	0.193069 ± 071	0.192637 ± 056	2764.8 ± 2.0
Weighted Mean age = 2764.4 ± 0.8 Ma, MSWD = 0.45						
SKS 21-4 (21.582°N; 85.971°E)						
SKS 21-4-1	25	24245	201	0.193025 ± 024	0.192527 ± 024	2763.9 ± 1.7
SKS 21-4-2	20	21492	630	0.193036 ± 027	0.192462 ± 036	2763.3 ± 1.8
SKS 21-4-3	25	19447	624	0.193194 ± 025	0.192562 ± 036	2764.2 ± 1.8
SKS 21-4-4	21	12618	387	0.193396 ± 056	0.192428 ± 057	2763.0 ± 2.0
SKS 21-4-5	13	10914	262	0.193516 ± 066	0.192402 ± 066	2762.8 ± 2.1
Weighted Mean age = 2763.5 ± 0.8 Ma, MSWD = 0.36						
K 9-3 (21.664°N; 85.874°E)						
K 9-3-1	18	49658	688	0.192678 ± 077	0.192413 ± 077	2762.9 ± 2.2
K 9-3-2	19	22056	1177	0.192956 ± 067	0.192415 ± 063	2762.9 ± 2.1
K 9-3-3	20	65724	1542	0.192539 ± 133	0.192358 ± 130	2762.4 ± 2.6
K 9-3-4	20	91207	2589	0.192588 ± 063	0.192449 ± 062	2763.2 ± 2.1
K 9-3-5	20	93611	2723	0.192663 ± 044	0.192529 ± 046	2763.9 ± 1.9
Weighted Mean age = 2763.2 ± 0.9 Ma, MSWD = 0.25						
SKS 13 (21.829°N; 85.594°E)						
SKS 13-1	29	22534	462	0.192689 ± 074	0.192140 ± 070	2760.6 ± 2.1
SKS 13-2	29	24057	275	0.192619 ± 046	0.192115 ± 046	2760.4 ± 1.9
SKS 13-3	29	22169	106	0.192543 ± 032	0.191997 ± 033	2759.3 ± 1.8
SKS 13-4	30	30947	782	0.192423 ± 026	0.192024 ± 034	2759.6 ± 1.8
SKS 13-5	27	28333	256	0.192463 ± 032	0.192038 ± 033	2759.7 ± 1.8
SKS 13-6	30	33311	275	0.192529 ± 029	0.192165 ± 030	2760.8 ± 1.8
SKS 13-7	25	17827	278	0.192714 ± 037	0.192035 ± 031	2759.7 ± 1.8
Weighted Mean age = 2760.0 ± 0.6 Ma, MSWD = 0.36						
K 24 (22.225°N; 86.079°E)						
K 24-1	30	45549	1019	0.192538 ± 083	0.192276 ± 071	2761.7 ± 2.2
K 24-2	21	25193	1306	0.192678 ± 125	0.192187 ± 131	2761.0 ± 2.6
K 24-3	27	11679	300	0.193266 ± 084	0.192225 ± 082	2761.3 ± 2.2
K 24-4	27	10293	246	0.193379 ± 089	0.192182 ± 085	2760.9 ± 2.2
K 24-5	30	18460	190	0.192192 ± 026	0.192136 ± 029	2760.5 ± 1.8
Weighted Mean age = 2761.0 ± 1.0 Ma, MSWD = 0.19						
SKS 27 (22.563°N; 85.892°E)						
SKS 27-1	30	16560	330	0.197719 ± 076	0.196990 ± 074	2801.4 ± 2.2
SKS 27-2	30	33433	968	0.197160 ± 034	0.196785 ± 036	2799.7 ± 1.9
SKS 27-3	30	33044	454	0.197109 ± 032	0.196747 ± 032	2799.4 ± 1.8
SKS 27-4	30	59877	1280	0.197057 ± 023	0.196852 ± 022	2800.3 ± 1.7
SKS 27-5	30	52162	1356	0.197110 ± 030	0.196877 ± 030	2800.5 ± 1.8
SKS 27-6	30	37587	402	0.197182 ± 042	0.196862 ± 042	2800.4 ± 1.9
Weighted Mean age = 2800.2 ± 0.7 Ma, MSWD = 0.49						
SKS 34 (22.609°N; 86.195°E)						
SKS 34-1	30	33288	317	0.191519 ± 017	0.191155 ± 019	2752.1 ± 1.7
SKS 34-2	19	17776	113	0.191833 ± 068	0.191143 ± 066	2752.0 ± 2.1
SKS 34-3	19	18082	261	0.191878 ± 078	0.191204 ± 078	2752.5 ± 2.1
SKS 34-4	19	16280	292	0.191771 ± 055	0.191033 ± 061	2751.1 ± 2.0
SKS 34-5	25	14990	115	0.191985 ± 090	0.191183 ± 089	2752.4 ± 2.2
Weighted Mean age = 2752.0 ± 0.9 Ma, MSWD = 0.30						

$^{206}Pb/^{204}Pb$ (m) and $^{207}Pb/^{206}Pb$ (m) are measured values (fraction means with standard error) and $^{207}Pb/^{206}Pb$ (c) is the corrected value. Uncertainties are $2\sigma_m$ and refer to the least significant digits. Age errors include fractionation uncertainty of 0.055%. Weighted mean ages were calculated using Ludwig (2003).

and are seen exposed over the entire craton. Individual dykes of this swarm can run for several tens of kilometres and vary in thickness from about 10 to 500 meters. Our sampling sites for both geochronology and paleomagnetism are confined to this swarm and are fairly evenly distributed covering almost the entire Singhbhum granite region, locations of all these sites have been given in (Fig. 1).

3. Results

3.1. Petrology

All the dykes sampled are doleritic in composition consisting of calcic plagioclase and clinopyroxene, generally in an ophitic texture. Mineral analysis of pyroxenes and plagioclase (Maity et al., 2008) show pyroxenes to vary in composition from augite, subcalcic augite, ferroaugite to pigeonite. Plagioclase varies in composition from albite to anorthite. Occasionally pyroxene is replaced by chlorite and actinolite. In parts of the swarm the fine grained plagioclase was subjected to different degrees of sericitization. In most samples along the dyke swarm plagioclase grains are clouded to varying degrees. Alteration of both pyroxene and plagioclase has been attributed to hydrothermal activity (Sengupta et al., 2014). Common accessory minerals in these rocks are magnetite associated occasionally with ilmenite. Microphotographs of samples typical of these dykes are shown in Fig. 2. Petrographic studies do not however show any major difference in the primary mineral compositions, other than the state of alteration either along strike or between these dyke swarms.

3.2. Baddeleyite geochronology

Results of Pb-Pb baddeleyite (ZrO_2) TE-TIMS analysis on eight NNE-SSW striking dykes from the Singhbhum craton are given in Table 1 and Figs. 3 and 4. Sample locations are given in Fig. 1 and their geographic coordinates in Table 1. TE-TIMS analysis (analytical procedures followed are given in Annexure-1) of at least five baddeleyite fractions each from six samples located in the south, central and northern parts of the dyke swarm yielded ages ranging between 2764 Ma and 2760 Ma. All the six age determinations overlap within errors suggesting simultaneous emplacement of these dykes within a brief time span of not more than 5 Ma (permitting maximum error limits), in virtually the entire Singhbhum granite region. The weighted mean age of these samples is 2762.4 ± 2.0 Ma which is inferred as the emplacement age of this swarm.

Samples SKS 27 and SKS 34 located on the northern fringe of the swarm have however yielded 2800.2 ± 0.7 Ma (MSWD: 0.49) and 2752.0 ± 0.9 Ma (MSWD: 0.30) ages respectively. Distinctly older and younger ages respectively compared to the six dyke age determinations given above. Barring a Neoarchean Rb-Sr isochron age of 2613 ± 177 Ma (Roy et al., 2004) on a NNE-SSW trending ultramafic dyke located (exact geographic coordinates of dyke not reported by the authors) west of site SKS 27, published K-Ar age determinations on these (newer dolerite) dykes are all Proterozoic in age varying from 2144 Ma to 950 Ma (Sarkar et al., 1969; Sarkar and Saha, 1977 and Mallik and Sarkar, 1994). These K–Ar whole-rock ages not only vary considerably between swarms but also within, this is therefore attributed to inherent limitations of the dating method when applied to Proterozoic rocks.

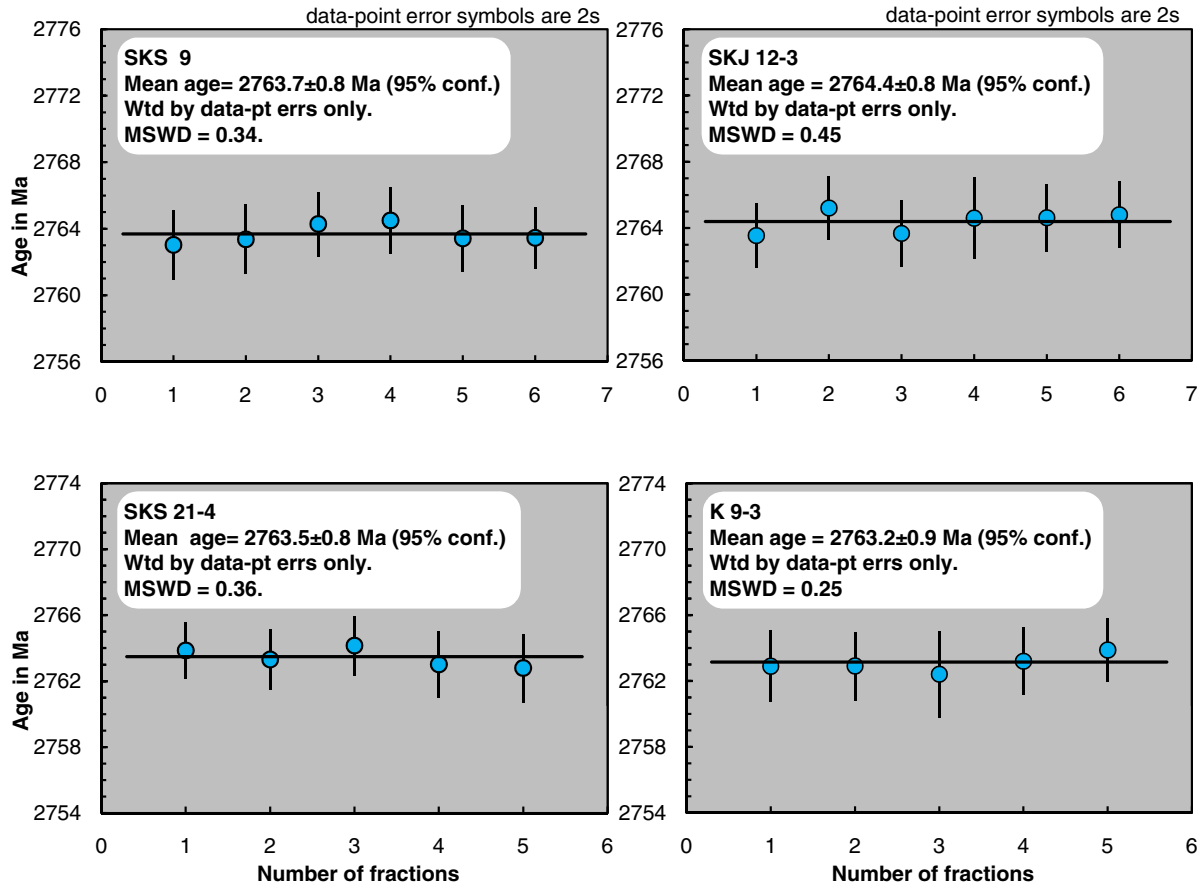


Fig. 3. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on baddeleyite fractions from samples SKS 9, SKJ 12-3, SKS 21-4, K 9-3. Error bars represent 95% confidence limits of measurements.

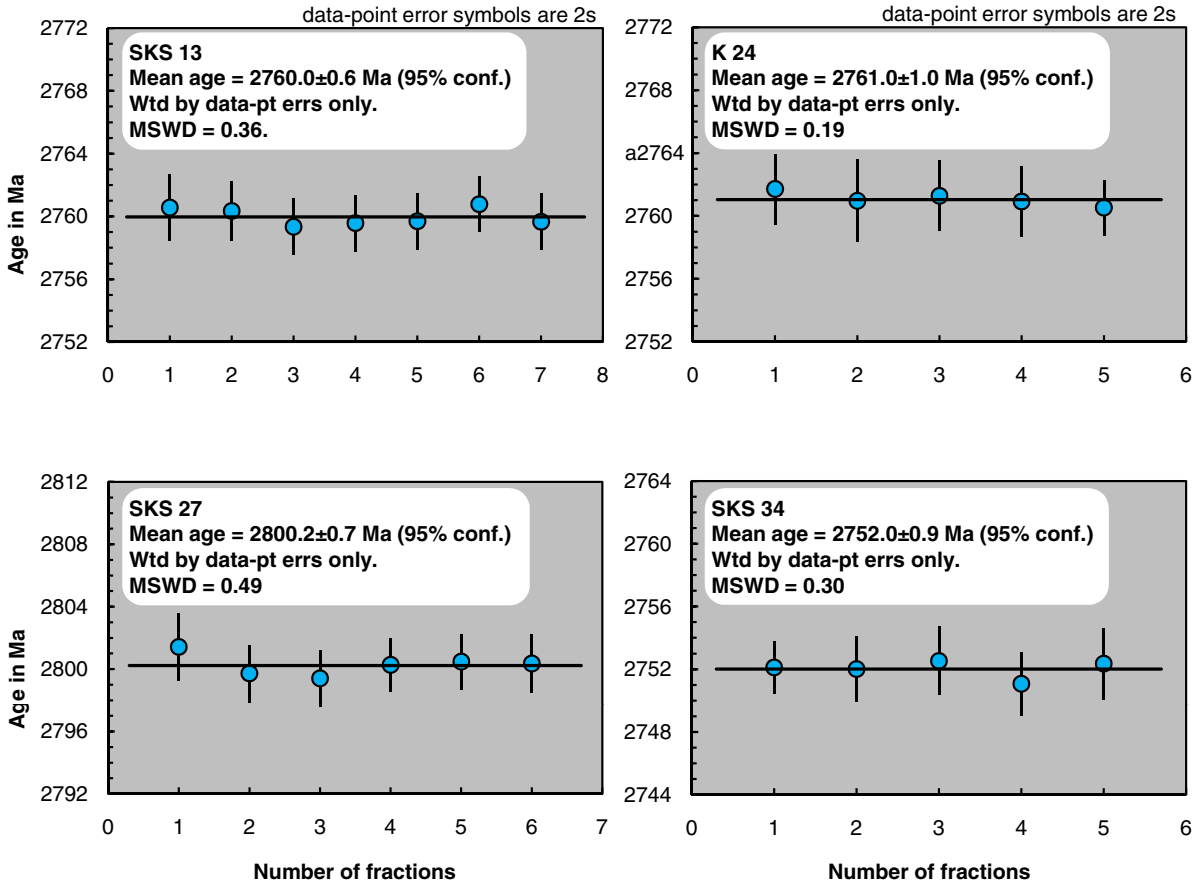


Fig. 4. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on baddeleyite fractions from samples SKS 13, K 24, SKS 27, SKS 34. Error bars represent 95% confidence limits of measurements.

Table 2
Paleomagnetic data of the NNE striking dykes from Singhbhum craton.

Site	Lat (°N)	Long (°E)	Strike (°N)	n/N	D (°)	I (°)	k	α_{95}	Plat (°N)	Plong (°E)	A_{95}
Normal dykes											
SKJ 12*	21.401	85.885	22	10	187	85	186	4	11	85	7
K 20	21.483	85.932	26	8	187	72	101	6	12	262	9
SKS 21*	21.582	85.971	14	10	201	78	166	4	0	78	7
K 2	21.573	85.994	22	10	318	85	76	6	29	78	11
K 3	21.577	85.989	25	10	264	85	264	3	20	75	6
K 4	21.579	85.975	10	10	191	83	205	3	9	84	7
K 13	21.615	85.762	15	8	200	68	66	7	15	253	11
K 26	22.146	86.235	36	9	287	82	31	9	26	70	18
K 24*	22.225	86.079	33	8	255	86	60	7	20	78	14
Q[a]	–	–	–	6	23	82	43	10	37	94	19
A[a]	–	–	–	7	92	86	85	6	22	94	12
N. dykes Mean			–	96/11	210	85	70	6	13	81	11
Reverse dykes											
K16	21.534	85.806	8	10	146	−84	94	5	31	78	10
K9*	21.664	85.874	40	10	113	−66	31	9	31	40	13
K7	21.684	85.854	13	10	11	−69	57	7	15	259	10
R. dykes Mean			–	30/3	79	−80	18	30	17	66	56
Grand Mean			–	126/14	226	84	47	6	14	78	11

Note: Lat(°N) and Long(°E) are site latitude and site longitude respectively; n/N is the number of samples from each site and number of sites contributing to the mean; S(°) is strike of the dyke at the respective sites; D(°) and I(°) are the mean declination and mean inclination; k is the Fisher's (1953) precision parameter of the mean pole; α_{95} is the radius of the circle of 95% confidence about the mean magnetization direction; Plat and Plong are the palaeolatitude and palaeolongitude for the paleomagnetic Poles; A_{95} = semi-angle of the 95% circle of confidence about the mean paleopole. * = sites with Pb-Pb geochronologic ages. Grand Mean was calculated after inverting the three reverse sites. [a]Verma and Prasad, 1974.

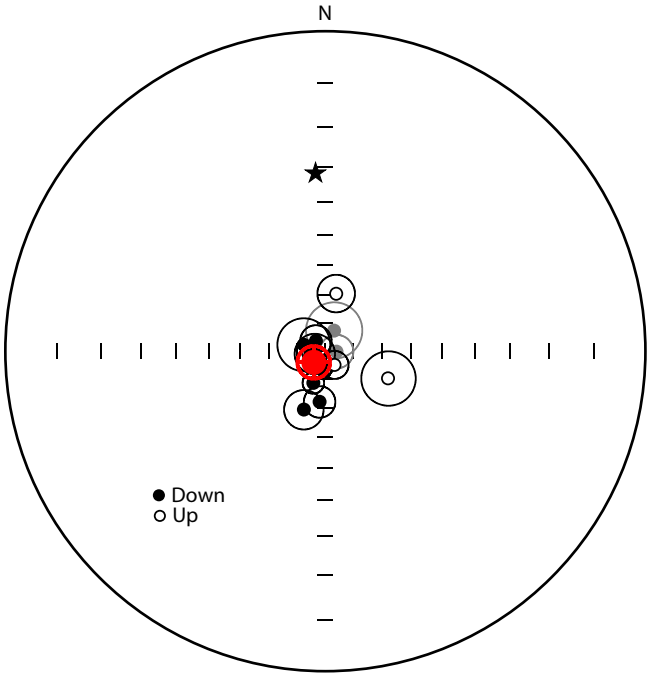


Fig. 5. Stereoplot showing paleomagnetic data. Site mean characteristic remanence directions with circles of 95% confidence of 12 sites from present study and 2 sites from previously published data. Present study site means are represented as black circles and published site means are represented as gray circles. Red closed circle represents the grand mean of present study and published study. Black star represent Present Earth's field (PEF) direction based on the 1995 IGRF. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Paleomagnetism

A total of 113 samples from 12 dykes (on 12 sites) were used for paleomagnetic investigation. Results are given in Table 2 and plotted in Figs. 5 and 6 (analytical procedures followed are given in Annexure-1). At least five samples from each site were subjected to detailed stepwise AF and/or thermal demagnetization, to a maximum of 150 mT or 600 °C, in order to separate the characteristic remanent magnetization component (ChRM,Fig. 6). A high coercivity (or high blocking temperature) component, probably recorded by magnetite, defines a well grouped characteristic magnetization direction, with southerly declination and steep-down inclination in nine sites and northerly steep-up in three sites (Table 2). Within errors, the nine normal and three reverse sites regardless of their geographic location in the dyke swarm have similar characteristic remanent magnetization directions of (D = 212°, I = 83°, α_{95} = 6°, N = 9) and (D = 79°, I = -80°, α_{95} = 30°, N = 3) respectively, (Table 2 and Fig. 5). Similar remanent magnetization directions were obtained by Verma and Prasad (1974) from the northern part of this dyke swarm at two sites, locations shown in Fig. 1. A reversal test (McFadden and McElhinny, 1990) performed on these samples was positive and classified as "Rc" with the observed angle γ = 7.7° and the critical angle γc = 14.3°. Though we were unable to obtain baked contact samples during this study to prove the primary nature of the remanence direction obtained. The preservation of dual polarity in this dyke swarm, and passage of the reversal test indicates that ChRM directions were not contaminated due to secondary magnetization (Buchan, 2013). Consistency in the remanent magnetization directions obtained across large geographic distances across and along the swarm, and importantly the younger (1765 Ma old, Shankar et al., 2014) dykes in this region preserve a distinctly different magnetization. Recently, Nelson et al. (2014) constrained the time of regional deformation and amphibolite grade metamorphism to the interval c. 3325–3300 Ma in the Singhbhum granite complex. This suggests that this dyke swarm could not have not been heated beyond their blocking temperature (~450–550 °C), after their emplacement. These evidences indicate that the nature of magnetization recorded by the dykes reported here was acquired at the time of their emplacement.

The 14 sites together yield a mean (after inverting the three reverse sites) of D = 226°, I = 84° (α_{95} = 6°) corresponding to a paleomagnetic pole of 14°N, 78°E (dp = 11, dm = 12).

4. Discussion

4.1. Neoarchean large igneous provinces

Based on the data obtained during this study the 2762.4 ± 2.0 Ma dyke swarm is the most prominent in the Singhbhum craton extending across the entire Singhbhum granite complex with an areal distribution of about 30,000 km^2. To assess the distribution of the older 2800.2 ± 0.7 Ma and younger 2752.0 ± 0.9 Ma dykes within this craton additional analysis on dykes from these swarms are required. It is likely that the NNE striking dykes which intrude the Bonai granite, on the western banks of the Singhbhum craton could also be coeval with one of these events. Though no ~2752 Ma events are known from this region, thus far, the older ~2800 Ma dyke event appears to be contemporaneous with several other magmatic events reported earlier from the Singhbhum craton (Fig. 1). The occurrence of dacite tuffs which yielded an igneous crystallization date of 2806 ± 6 Ma (U-Pb zircon) within the Malaigiri basin (Nelson et al., 2014), on the south-western margin of the craton, the Budhapal granite dated at 2807 ± 13 Ma (U-Pb zircon, Bose et al., 2016), Rengali felsic gneiss, hornblend gneiss and charnokite emplacement ages of 2776 ± 24 Ma, 2829 ± 9 Ma, 2861 ± 30 Ma respectively (Bose et al., 2016) with the Rengali granite having an age of 2803 ± 4 Ma (U-Pb zircon) from the south-western and 2811 ± 3 Ma (U-Pb zircon) from southern parts of the craton (Misra et al., 1999). The Temperkola granite activity at 2809 ± 12 Ma (U-Pb zircon) and the emplacement of associated acid volcanics at 2822 ± 67 Ma (Bandyopadhyay et al., 2001) located on the western periphery of the craton, and perhaps (with an imprecise U-Pb zircon age of 2837 ± 200 Ma, Acharyya et al., 2010) the Mayurbhanj granite (fine-grained phase) are also contemporaneous. Pb-Pb and Sm-Nd whole rock ages (Misra and Johnson, 2005), of 2794 ± 210 Ma and 2787 ± 270 Ma respectively, though with large errors also indicate a ball park age of ~2800 Ma for the Dhanjori volcanics (Fig. 1). This suggests a fairly wide spread thermal event (largely confined to the southern and to some extent northern periphery of the Singhbhum granite) at ~2800 Ma.

Though Mesoarchean and Neoarchean mafic magmatism has been reported from several other continents, dyke swarms of such antiquity were unknown from any of the cratons within the Indian subcontinent until this report. The oldest known is the Paleoproterozoic ~2367 Ma Dharwar giant dyke swarm, in the eastern Dharwar craton (Halls et al., 2007 and Kumar et al., 2012). However, igneous provinces contemporaneous with the Neoarchean dykes reported here are known from the Pilbara craton. Magmatic events within the Nullagine Supersequence belonging to Packages 2 and 3 were dated at 2766 ± 2 Ma (Dacite) and 2766 ± 3 Ma (Rhyolite) respectively (Blake et al., 2004). The Gregory granite complex with crystallization (Zircon SHRIMP) ages of 2761 ± 2 Ma (Nelson, 1999), 2762 ± 4 Ma and 2763 ± 8 Ma (Nelson, 1997), and the Pre-Fortescue Group granitic rocks (Whim Creek Belt) with an age (zircon) of 2765 ± 5 Ma (Nelson, 1997), are all coeval with this event.

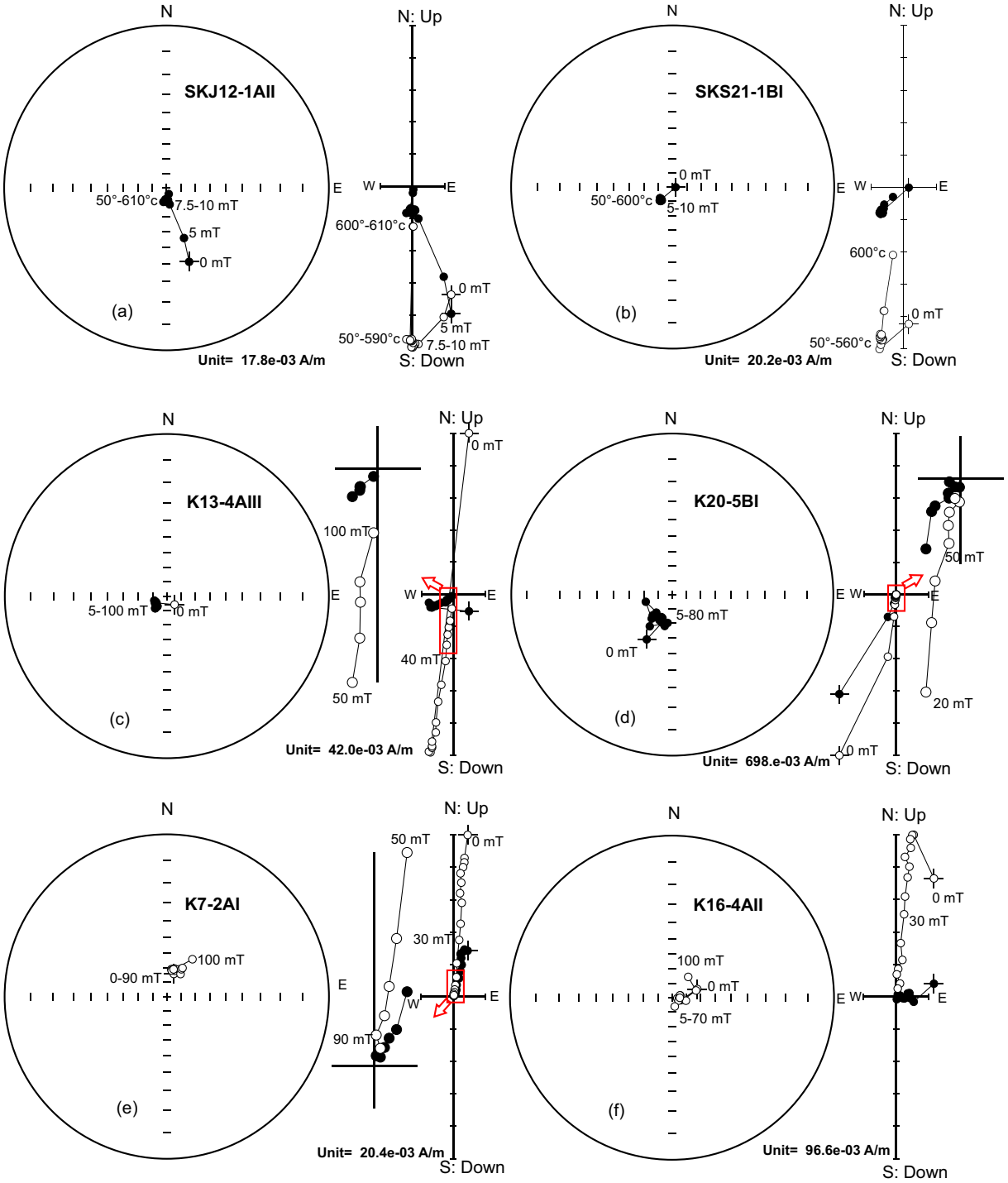


Fig. 6. Zijderveld diagrams and equal area stereonet projections showing characteristic behaviour of natural remanence to AF and thermal demagnetization (a and b) and AF demagnetization (c–f) for representative samples from different sites. Thermal demagnetization measurements are in °C and AF measurements are in millitesla (mT). Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda, 2006).

While felsic tuffs in the Package 4 of the Nullagine Supersequence yielding an eruption and deposition age of 2752 ± 5 Ma (Blake et al., 2004), is coeval with the younger dyke event (2752.0 ± 0.9 Ma) reported here. The Sylvania mafic dykes (2747 ± 4 Ma, Wingate, 1999) from the eastern Pilbara, felsic volcanism belonging to the Harley Formation (Southern Pilbara region), a granite precursor in the Gregory area dated at 2750 ± 5 Ma (Hall, 2005) and 2757 ± 5 Ma (Nelson, 1997) respectively are all coeval with this event. On the other hand, the Fortescue mafic magmatic event comprising of the Black range dykes is however, older with an emplacement age of 2772 ± 2 Ma (Wingate, 1999). Strictly, none of the three dyke events reported here are yet known from the Kaapvaal craton, however, the Derdepoort outlier with an emplacement age of 2769 ± 2 Ma (Walraven et al., 1996) located between the Kanye and the Ventersdorp proper outcrops, east of Gaborone is an event temporally close to the dominant dyke event (2762 ± 2 Ma) reported here. The Kanye Volcanic Formation with an age of 2780 ± 2 Ma (Walraven et al.,

1996), the 2782 ± 5 Ma old Derdepoort basalt (Wingate, 1998) and the Modipe Gabbro (2784 ± 1 Ma, Denyszyn et al., 2013) are all older than the 2762 event but significantly younger than the older dyke event (2800 Ma) reported from the Singhbhum.

4.2. Paleoreconstruction of Singhbhum craton with respect to Vaalbara supercontinent

To verify if the igneous provinces, mentioned above, were discrete events or they were spreading over the three cratons Singhbhum, Pilbara and Kaapvaal that may have been neighbours during the Neoarchean, we compared their paleolatitudinal positions. A similar study comparing the paleopositions of Kaapvaal and Pilbara during the same time span was attempted by Denyszyn et al. (2013). Data from the Modipe gabbro (with an age of 2784 ± 1 Ma, Denyszyn et al., 2013) and the Derdepoort basalts (2782 ± 5 Ma, Wingate, 1998) which yielded paleolatitudes of 64.5° ± 8.6° (after tilt-correction) and 64.5 ± 17.5° (Wingate, 1998) respectively, both from Kaapvaal were compared with the Mount Roe Basalts (coeval with the Blake range suite dated at 2772 ± 2 Ma, Wingate, 1999) that provided a paleolatitude of 51.1° ± 3.9° (Strik et al., 2003), from the Pilbara. Based on this data the authors (Denyszyn et al., 2013) concluded that paleolatitudinal positions of these two cratons overlapped within the quoted uncertainties and supported the Vaalbara hypothesis. Paleomagnetic data (this study) from the Singhbhum Neoarchean dyke swarm shows this craton was located at a paleolatitude of 79° ± 12° at 2762 ± 2 Ma. Coeval magmatic event from the Pilbara dated at 2766 ± 2 Ma (Blake et al., 2004) within the Nullagine Supersequence belonging to Package 2 shows this craton was at a paleolatitude of 53.6 ± 15° (Strik et al., 2003) and package 4 of the same Supersequence though marginally younger at 2752 ± 5 Ma (Blake et al., 2004) also indicates Pilbara was positioned at a moderately steep paleolatitude of 62 ± 9° (Strik et al., 2003). As mentioned earlier the Mount Roe Basalts (Strik et al., 2003), from the Pilbara show that this craton was located at moderately steep latitudes, though not as steep as the Singhbhum and Kaapvaal, during that time. Despite the limitations (due to unknown drift rates of these cratons during that time, and that the sampled sites could be 180 degrees apart if alternate polarity is used) of comparing paleomagnetic data recorded by formations about 20 Ma apart (Modipe Gabbro and Derdepoort basalts), the possibility that Singhbhum could have been contiguous with Kaapvaal and Pilbara (in Vaalbara) cannot be ruled out.

Previous studies, based on the stratigraphic analogy found between the Kaapvaal and Pilbara cratons a supercraton Vaalbara was proposed (Cheney, 1996; Zegers et al., 1998). This hypothesis was validated by the reconstruction of paleopositions of Kaapvaal and Pilbara cratons and the alignment of tectonic lineaments across them (Zegers et al., 1998; Strik et al., 2003; de Kock et al., 2009; Denyszyn et al., 2013).

Encouraged by the overlap of paleolatitude data from the Singhbhum and Kaapvaal a paleoreconstruction was attempted (data used is listed in Table 3, Fig. 7), to evaluate the proximity of the Singhbhum craton with respect to both Kaapvaal and Pilbara. Adopting a similar approach to that used by de Kock et al. (2009), their chosen Euler rotation pole of λ = 59.0°S, φ = 251.5°E and an angle of 93.2° was used to rotate Pilbara poles. On doing this, poles from the two cratons overlap and place Pilbara to the north-west of the present-day Kaapvaal craton in a Kaapvaal reference frame. In this configuration several Mesoarchean and Paleoproterocoic features appear to be contiguous across the two cratons (de Kock et al., 2009). Subsequently, Singhbhum pole was rotated by 99.2° about a Euler pole (λ = 38.2°S, φ = 117.6°E) to superpose it with the mean of five Pilbara poles (with < 10 Ma age difference compared to the Singhbhum dykes) Package 2, Package 1, Blake Range dykes, Cajuput Dykes and Mount Roe basalts (listed in Table 3) to obtain its position south of the Kaapvaal craton. Allowing for errors on the Singhbhum and the mean Pilbara pole data, position of the Sinbhbhum craton can be moved closer to the Kaapvaal using a Euler pole of λ = 38.7°S, φ = 114.0°E with an angle of 91.2°. Though several other possibilities of reconstructing Sinbhbhum with respect to Kaapvaal are possible, our main objective here was to show that Singhbhum could have been located adjacent to Kaapvaal during that time (~2770 Ma ago).

4.3. Development of the Singhbhum with Pilbara and Kaapvaal cratons during the Archean

Comparative studies of the Archean geological history of Singhbhum craton with the Kaapvaal (Hofmann et al., 2016) and Pilbara (Nelson et al., 2014) cratons, have revealed several similarities between the three cratons. The oldest rocks reported from the Singhbhum craton, felsic volcaniclastic rocks from the Daitari

Table 3
Poles used for Paleogeographic reconstruction.

Name	Age (Ma)	P_{lat} (°N)	P_{long} (°E)	A_{95}	N	References
Pilbara craton						
Medium temperature pole from Package 7–10 Basalt (MT7–10)	2715–2724	−53.2	203.9	5.3	12	Strik et al. (2003), Blake et al. (2004)
Package 8–10 Basalt (P8-10)	2718–2715	−59.1	186.3	6.1	9	Strik et al., 2003, Blake et al. (2004)
Package 4–7 Basalt (P4-7)	2752–2721	−50.4	138.2	12.5	8	Strik et al. (2003), Blake et al. (2004)
Package 2 Basalt (P2)	2766 ± 2	−46.5	152.7	15.2	3	Strik et al. (2003), Blake et al. (2004)
Package 1 Basalt (P1)	2772 ± 2	−40.8	159.8	3.7	13	Strik et al. (2003), Blake et al. (2004)
Black Range Dykes (BR)	2772 ± 2	−32.0	154.0	9.0	16	Embleton (1978), Wingate (1999)
Cajuput Dykes (CD)	2772 ± 2	−46.0	146.0	22.0	9	Embleton (1978), Wingate (1999)
Mount Roe Basalt (RB)	2775–2763	−52.4	178.0	7.6	12	Schmidt and Embleton (1985), Arndt et al. (1991)
Kaapvaal craton						
Allanridge Formation (DC)	2718–2713	−69.8	345.6	5.8	18	de Kock et al. (2009)
Derdepoort Basalt (DB)	2782 ± 5	−39.6	4.7	17.5	8	Wingate (1998)
Modipe Gabbro (MG)	2784 ± 1	−32.8	30.9	10.5	10	Evans and McElhinny (1966), Denyszyn et al. (2013)
Modipe Gabbro (DZ)	2784 ± 1	−47.6	12.4	8.9	11	Denyszyn et al. (2013)
Singhbhum craton						
NNE Dykes (SI)	2762 ± 2	14.0	78.0	11	14	Present study

Name: Name of pole, Age: Radiometric/Stratigraphic age of pole in Ma, P_{lat} and P_{long} are Pole latitude (°N) and Pole longitude (°E) of the Paleopoles; A_{95}: ellipses of confidence at the 95% level about the mean paleopole, N: number of sites studied, References given are for paleomagnetic data and age determination (in parenthesis). Only reliably dated poles are listed above and used in the reconstruction.

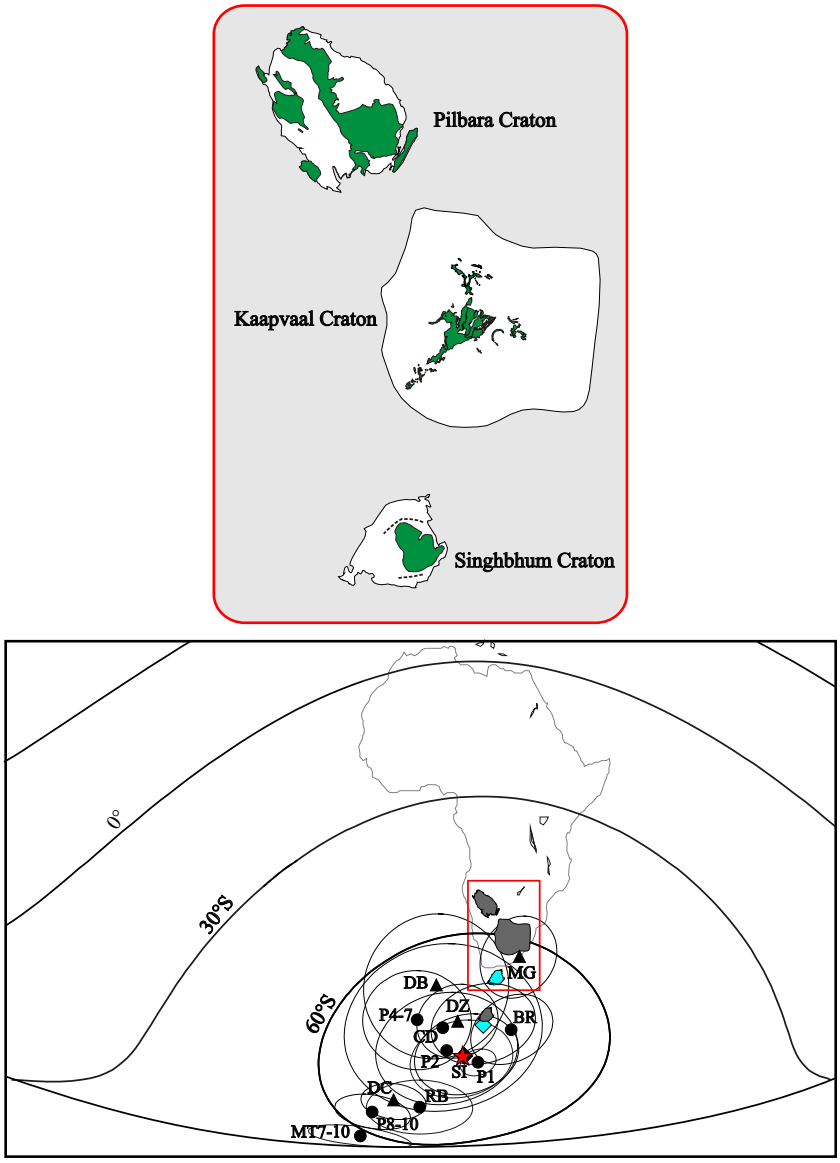


Fig. 7. Mollweide projection showing paleopositions of Kaapvaal, Pilbara and Singhbhum cratons, in light salmon pink shade, during the Neoarchean (~2770 Ma). Black triangles are Kaapvaal poles, black dots represent Pilbara poles and black square is the Singhbhum pole. Pilbara poles were rotated to overlap with the Kaapvaal poles with the Euler pole at λ = 59.0°S, ϕ = 251.5°E by 93.2° (de Kock et al., 2009) and the Singhbhum pole around a pole at λ = 38.2°S, ϕ = 117.6°E with an angle of 99.2°. This reconstruction was done in a Kaapvaal reference frame. Star represents the mean of five Pilbara poles (P2, P1, BR, CD and RB). Given in blue is the alternate location of the Singhbhum pole and craton allowing for errors on the Singhbhum and Pilbara pole data. This reconstruction was done using "PaleoMac" computer program (Cogne, 2003). Inset is an enlarged view of the reconstructed cratons showing the 2.70–2.775 Ga mafic magmatism in the Pilbara, 2.70–2.785 Ga mamatism in the Kaapvaal and 2.762 Ma magmatism in the Singhbhum cratons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

greenstone belt which (Iron Ore Group) have been dated at 3507 ± 2 Ma (Mukhopadhyay et al., 2008) are equated with the lower portions of the Onverwacht (between 3.53 and >3.43 Ga, Taylor et al., 2016) and Nondweni groups (3.53–3.41 Ga by U-Pb on zircon; Xie et al., 2012) of the Kaapvaal craton. These are also contemporaneous with the so far known oldest greenstone sequence from the eastern Pilbara craton, namely the 3515 ± 3 Ma Coonterunah succession (Buick et al., 1995). It has been observed that granite emplacement reached a maximum in the Sinbhbhum craton during regional metamorphism between 3325 and 3300 Ma (Acharyya et al., 2012; Nelson et al., 2014; Upadhyay et al., 2014), synchronous with the emplacement of the major granitoid complexes in the Eastern Pilbara Terrane between ~ 3325 and 3315 Ma (Nelson, 1999). Most importantly, early stabilization of both the central Singhbhum and Kaapvaal cratons that was marked by the wide spread emplacement of the 3.1 Ga granites (Poujol et al., 2003; Eglington and Armstrong, 2004; Schmitz et al., 2004; Schoene et al., 2008) in the Kaapvaal are coeval with the youngest suite of the TTG and granite assemblage in the Singhbhum (Saha, 1994). By and large cratonization in the Pilbara also was complete by about 3.0 Ga (Moser et al., 2001), however, it has been shown that final cratonization was effected by the emplacement of 2.89 to 2.83 Ga post-tectonic granites (Van Kranendonk et al., 2007). Intrusion of these granites along a NW-SE trending corridor across the Kurrana Terrane-Eastern Pilbara boundary was inferred to represent a failed rift (Van Kranendonk et al., 2007). Similar granite emplacement between 2.803 and 2.861 Ga (Misra et al., 1999; Bandyopadhyay et al., 2001; Nelson et al., 2014; Bose et al., 2016), largely concentrated to the south of the Singhbhum granite along a WNW-ESE trending stretch (Fig. 1), could be the response of a similar tectonic event (rift) in the Singhbhum. Several similarities have been observed in the lithostratigraphy (Hofmann et al., 2016) of the coeval continental cover sediments of the western Iron Ore Group of the

Singhbhum craton and the Pongola Supergroup (2.99–2.87 Ga, Gumsley et al., 2015) and Witwatersrand basin sediments (2.99–2.78 Ga, Kositcin et al., 2003; Kositcin and Krapez, 2004) in the Kaapvaal. Notably, the shear zones that encircle the granite complexes of the Eastern Pilbara Terrane were considered (Nelson et al., 2014) to be similar to the Singhbhum shear zone (200 km long and 10 km wide, Bhattacharya et al., 2014) which forms the craton's north boundary and the Sukinda thrust that restricts it in its southern extent.

Following the super-craton hypothesis, wherein at least three 'clans' of cratons, a Kaapvaal craton-like clan, a Superior craton-like clan and a Slave craton-like clan, have been suggested (Bleeker, 2003). Each clan with characteristically distinct geological histories can be traced to their ancestral super-craton. The essential differences between these super-cratons include their age of cratonization, chronology of volcano-sedimentary cratonic cover successions and the time of mafic dyke swarm emplacement (Bleeker, 2003).

The Kaapvaal like clan which include the Kaapvaal and Pilbara cratons with Vaalbara super-craton being their progenitor (Cheney, 1996), cartonized by ~3.0 Ga (Moser et al., 2001). As discussed above, the Singhbhum apparently cratonized at about the same time (Roy and Bhattacharya, 2012), much before either the Slave or Superior cratons which had cratonized subsequent to the granite magmatism ("granite bloom") during 2600–2580 Ma (van Breemen et al., 1992; Davis and Bleeker, 1999; Davis et al., 2003) or granite plutonism between 2680 and 2649 Ma (Card, 1990) respectively. Further, coeval deposition (after 3.0 Ga, following cratonization) of the western Iron Ore Group in the Singhbhum and the Pongola and Witwatersrand basin, both consisting of similar packages of shallow-marine quartzite, continental flood basalts and a ferruginous/manganiferous shale-BIF, suggests a good correspondence between the Singhbhum and Kaapvaal cratons. Finally, the large dyke swarms, now known from the Singhbhum (2762 ± 2 Ma, 2752.0 ± 0.9 Ma and 2800.2 ± 0.7 Ma, this study) are a clear expression of the characteristic Neoarchean mafic magamatism reported from both the Kaapvaal and Pilbara cratons. These comparisons show significant similarity in the Archean geological history of the Singhbhum with Kaapvaal and Pilbara cratons which belong to the Kaapvaal clan, suggesting Singhbhum craton to be a part of the Vaalbara supercraton.

5. Conclusions

Precise geochronology combined with paleomagnetism provided evidence for the discovery of hitherto unknown Neoarchean dyke swarms emplaced at 2762.4 ± 2.0 Ma, 2800.2 ± 0.7 Ma and 2752.0 ± 0.9 Ma within the Singhbhum craton. The extent of the 2762.4 ± 2.0 Ma dyke event has been established to occur over the entire Singhbhum craton with an areal distribution of about 30,000 km^2, suggesting it could be a remnant of a large igneous province extending far beyond the Singhbhum craton.

The older 2800.2 ± 0.7 Ma event is coeval with volcanic and granitic activity in the Singhbhum as well as in the Pilbara craton, suggesting this could also be a fairly wide spread thermal or tectonic event at that time.

Reasonably similar Archean geology and an acceptable supercraton fit of the Singhbhum, with Kaapvaal and Pilbara cratons during the Neoarchean, suggests Singhbhum could also have had its ancestry in the supercraton Vaalbara.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2017.01.018.

References

Acharyya, S.K., Gupta, A., Orihashi, Y., 2010. Neoarchaean-Palaeoproterozoic stratigraphy of the Dhanjori basin, Singhbhum Craton, Eastern India: and recording of a few U-Pb zircon dates from its basal part. J. Asian Earth Sci. 39, 527–536.

Acharyya, S.K., Gupta, A., Orihashi, Y., 2012. New U-Pb zircon ages from Paleo-mesoarchean TTG gneisses of the Singhbhum Craton, Eastern India. Geochem. J. 44, 81–88.

Arndt, N.T., Nelson, D.R., Compston, W., Trendall, A.F., Thorne, A.M., 1991. The age of the Fortescue Group, Hamersley Basin, Western Australia, from ion microprobe zircon U-Pb results. Aust. J. Earth Sci. 38 (3), 261–281.

Bandyopadhyay, P.K., Chakrabarti, A.K., DeoMurari, M.P., Misra, S., 2001. 2.8 Ga old anorogenic granite-acid volcanics association from western margin of the Singhbhum-Orissa craton, eastern India. Gondwana Res. 4, 465–475.

Bayanova, T.B., 2006. Baddeleyite: a promising geochronometer for alkaline and basic magmatism. Petrologiya 14, 203–216.

Bhattacharya, H.N., Nelson, D.R., Thern, E.R., Altermann, W., 2014. Petrogenesis and geochronology of the Arkasani Granophyre and felsic Dalma volcanic rocks: implications for the evolution of the Proterozoic North Singhbhum Mobile Belt, east India. Geol. Mag., http://dx.doi.org/10.1017/S0016756814000442 (submitted for publication).

Blake, T.S., Buick, R., Brown, S.J.A., Barley, M.E., 2004. Geochronology of a Late Archaean flood basalt province in the Pilbara Craton, Australia: constraints on basin evolution, volcanic and sedimentary accumulation, and continental drift rates. Precambr. Res. 133, 143–173.

Bleeker, W., 2003. The late Archean record: a puzzle in ca. 35 pieces. Lithos 71, 99–134.

Bose, M.K., Goles, G.G., 1970. Chemical petrology of the ultramafic minor intrusions of Singhbhum, Bihar. Proc. Sec. Symp. Upper mantle project. Hyderabad, 305, 326.

Bose, S., Das, K., Kimura, K., Hidaka, H., Dasgupta, A., Ghosh, G., Mukhopadhyay, 2016. Neoarchean tectonothermal imprints in the Rengali Province, eastern India and their implication on the growth of Singhbhum Craton: evidence from zircon U–Pb SHRIMP data. J. Metamorphic Geol. 2016 (34), 743–764.

Buchan, 2013. Key paleomagnetic poles and their use in Proterozoic continent and supercontinent reconstructions: a review. Precamb. Res. 244, 5–22.

Buick, R., Thornett, J.R., McNaughton, N.J., Smith, J.B., Barley, M.E., Savage, M., 1995. Record of emergent continental crust ~3.5 billion years ago in the Pilbara craton of Australia. Nature 375, 574–577.

Card, K.D., 1990. A review of the Superior Province of the Canadian Shield, a product of Archean accretion. Precambr. Res. 48, 99–156.

Chadima, M., Hrouda, F., 2006. Remasoft 3.0 user-friendly data browser and analyzer. Travaux Geophys. XXVII, 20–21.

Cheney, E.S., 1996. Sequence stratigraphy and plate tectonic significance of the Transvaal succession of southern Africa and its equivalent in Western Australia. Precambr. Res. 79, 3–24.

Cogne, J.P., 2003. Paleo Mac: a MacintoshTM application for treating paleomagnetic data and making plate reconstructions. Geochem. Geophys. Geosyst. 4, 1007. http://dx.doi.org/10.1029/2001GC000227.

Davis, W.J., Bleeker, W., 1999. Timing of plutonism, deformation, and metamorphism in the Yellowknife Domain, Slave Province, Canada. Can. J. Earth Sci. 36, 1169–1187.

Davis, W.J., Jones, A.G., Bleeker, W., Grü tter, H., 2003. Lithosphere development in the Slave craton: a linked crustal and mantle perspective. Lithos 71, 575–589.

de Kock, M.O., Evans, D.A., Beukes, N.J., 2009. Validating the existence of Vaalbara in the Neoarchean. Precambr. Res. 174, 145–154.

Denyszyn, S.W., Feinberg, J.M., Renne, P.R., Scott, G.R., 2013. Revisiting the age and paleomagnetism of the Modipe Gabbro of South Africa. Precambr. Res. 238, 176–185.

Eglington, B.M., Armstrong, R.A., 2004. The Kaapvaal Craton and adjacent orogens, southern Africa: a geochronological database and overview of the geologicaldevelopment of the craton. S. Afr. J. Geol. 107, 13–32.

Embleton, B.J.J., 1978. The paleomagnetism of 2400 M.Y. old rocks from the Australian Pilbara Craton and its relation to Archean-Proterozoic tectonics. Precambr. Res. 6, 275–291.

Evans, M.E., McElhinny, M.W., 1966. The paleomagnetism of the Modipe Gabbro. J. Geophys. Res. 71 (24), 6053–6063.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A Math. Phys. Sci. 217, 295–305.

Gumsley, A., Olssona, J., Söderlund, U., de Kock, M., Axel Hofmann, A., Klausend, M., 2015. Precise U-Pb baddeleyite age dating of the Usushwana Complex, southern

Africa – implications for the Mesoarchaean magmatic and sedimentological evolution of the Pongola Supergroup, Kaapvaal Craton. Precambr. Res. 267, 174–185.
Hall, C.E., 2005. SHRIMP U-Pb depositional age for the lower Hardey Formation: evidence for diachronous deposition of the lower Fortesque Group in the southern Pilbara region, Western Australia. Aust. J. Earth Sci. 52, 403–410.
Halls, H.C., Kumar, A., Srinivasan, R., Hamilton, M.A., 2007. Paleomagnetism and U-Pb geochronology of easterly trending dykes in the Dharwar craton, India: feldspar clouding, radiating dyke swarms and the position of India at 2.37 Ga. Precambr. Res. 155, 47–68.
Heaman, L.M., LeCheminant, A.N., 1993. Paragenesis and U-Pb systematics of baddeleyite (ZrO2). Chem. Geol. 110, 95–126.
Heaman, L.M., Tarney, J., 1989. U-Pb baddeleyite ages for the Scourie dyke swarm, Scotland: evidence for two distinct intrusion events. Nature 340, 705–708.
Hofmann, A., Jodder, J. and Xie, H. 2016. On the remarkable similarity of the Archaean geological evolution of the Singhbhum and Kaapvaal cratons. Abstract, 35th Inter. Geol. Cong. 27th Aug to 4th Sept. 2016, Cape Town South Africa.
Taylor, J., Zeh, A., Gerdes, A., 2016. U-Pb–Hf isotope systematics of detrital zircons in high grade paragneisses of the Ancient Gneiss Complex, Swaziland: Evidence for two periods of juvenile crust formation, Paleo- and Mesoarchaean sediment deposition, and 3.23 Ga terrane accretion. Precambr. Res. 280, 205–220.
Kositcin, N., Krapez, B., 2004. SHRIMP U-Pb detrital zircon geochronology of the Late Archaean Witwatersrand Basin of South Africa: relation between zircon provenance age spectra and basin evolution. Precambr. Res. 129, 141–168.
Kositcin, N., McNaughton, N.J., Griffin, B.J., Fletcher, I.R., Groves, D.I., Rasmussen, B., 2003. Textural and geochemical discrimination between xenotime of different origin in the Archaean Witwatersrand Basin, South Africa. Geochim. Cosmochim. Acta 67, 709–731.
Kumar, A., Hamilton, M.A., Halls, H.C., 2012. A Paleoproterozoic giant radiating dyke swarm in the Dharwar craton, southern India. Geochem. Geophys. Geosyst. 13. http://dx.doi.org/10.1029/2011GC003926.
Kumar, A., Nagaraju, E., Srinivasa Sarma, D., Davis, D.W., 2014. Precise Pb baddeleyite geochronology by the thermal extraction-thermal ionization mass spectrometry method. Chem. Geol. 372, 72–79.
Li, X.H., Su, L., Chung, S.L., Li, Z.X., Liu, Y., Song, B., Liu, D.Y., 2005. Formation of the Jinchuan ultramafic intrusion and the world's third largest Ni–Cu sulfide deposit: associated with the 825 Ma south China mantle plume? Geochem. Geophys. Geosyst. 6, Q11004. http://dx.doi.org/10.1029/2005GC001006.
Ludwig, K. R., 2003. User's Manual for Isoplot 3.00: A Geochronological Toolkit for Excel: Berkeley Geochronological Center Special Publication 4, pp. 71.
Maity, B. K., Ray, J. S., Chattopadhyay, B., Sengupta, S., Nandy, S. Saha, A., 2008. Contrasting petrological variants in newer dolerite dyke swarm around Bisoi, Eastern Indian Shield: Insights from petrography and mineral chemistry. In: Indian Dykes: Geochemistry, Geophysics and Geochronology, pp. 447–470.
Mallik, A.K., Sarkar, A., 1994. Geochronology and geochemistry of mafic dykes from the Precambrians of Keonjhar, Orissa. Ind. Minerals 48, 13–24.
McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int. 103, 725–729.
Misra, S., Johnson, P.T., 2005. Geochronological constraints on evolution of Singhbhum mobile belt and associated basic volcanics of eastern Indian shield. Gondwana Res. 8, 129–142.
Misra, S., Moitra, S., Bhattacharya, S., Shivaraman, T.V., 1999. Archaean granitoids at the contact of Eastern Ghats Granulite Belt and Singhbhum-Orissa Craton in Bhuban-Rengali sector, Orissa, India. Gondwana Res. 3, 205–213.
Moser, D.E., Flowers, R.M., Hart, R., 2001. Birth of the Kaapvaal Tectosphere 3.08 Billion Years Ago. Science 291, 465–468.
Mukhopadhyay, J., Beukes, N.J., Armstrong, R.A., Zimmermann, U., Ghosh, G., Medda, R.A., 2008. Dating the oldest greenstone in India: a 3.51 Ga precise U-Pb SHRIMP zircon age for dacitic lava of the Southern Iron Ore Group. Singhbhum Craton. J. Geol. 116, 449–461.
Nelson, D.R., 1997. Compilation of SHRIMP U–Pb zircon geochronology data, 1996. Geological Survey of Western Australia, Record 1997/2, pp.189.
Nelson, D.R., 1999. Compilation of SHRIMP U–Pb zircon geochronology data, 1998. Geological Survey of Western Australia, Record 1999/2, pp.222
Nelson, D.R., Bhattacharya, H.N., Thern, E.R., Altermann, W., 2014. Geochemical and ion-microprobe U-Pb zircon constraints on the Archaean evolution of Singhbhum Craton, eastern India. Precambr. Res. 255, 412–432.
Poujol, M., Robb, L.J., Anhaeusser, C.R., Gericke, B., 2003. A review of the geochronological constraints on the evolution of the Kaapvaal Craton, South Africa. Precambr. Res. 127, 181–213.
Roy, A.B., Bhattacharya, H.N., 2012. Tectonic and stratigraphic reappraisal constraining the growth and evolution of the Singhbhum Craton, eastern India. J. Geol. Soc. India 80, 455–469.
Roy, A., Sarkar, A., Jeyakumar, S., Aggarwal, S. K., Ebihara, M., Satoh, H., 2004. Late Archaean mantle metasomatism below eastern Indian craton: Evidence from trace elements, REE geochemistry and Sr-Nd-O isotope systematics of ultramafic dykes. In Proc. Indian Acad. Sci. (Earth Planet. Sci.). 113, 649–665.
Saha, A.K., 1994. Crustal evolution of north Singhbhum-Orissa, eastern India. Geol. Soc. India Memoir 27, 341.
Saha, A.K., Sankaran, A.V., Bhattacharyya, T.K., 1973. Geochemistry of the Newer dolerite suite of intrusions within the Singhbhum Granite. J. Geol. Soc. India 14, 324–346.
Sarkar, S.N., Saha, A.K., 1977. The present status of the Precambrian stratigraphy, tectonics and geochronology of Singhbhum-Keonjhar-Mayurbhanj region, eastern India. Indian J. Earth Sci. 4, 37–65.
Sarkar, S.N., Saha, A.K., Miller, J.A., 1969. Geochronology of the Pre-Cambrian rocks of Singhbhum and adjacent regions, Eastern India. Geol. Mag. 106, 15–45.
Schmidt, P.W., Embleton, J.J., 1985. Prefolding and overprint magnetic signatures in Precambrian (2.9–2.7 Ga) igneous rocks from the Pilbara Craton and Hamersley Basin, NW Australia. J. Geophys. Res. 90 (B4), 2967–2984.
Schmitz, M.D., Bowring, S.A., de Wit, M.J., Gartz, V., 2004. Subduction and terranecollision stabilized the western Kaapvaal craton tectosphere 2.9 billion yearsago. Earth Planet. Sci. Lett. 222, 363–376.
Schoene, B., de Wit, M.J., Bowring, S.A., 2008. Mesoarchean assembly and stabilization of the eastern Kaapvaal craton: a structural-thermochronologicalperspective. Tectonics 27, TC5010. http://dx.doi.org/10.1029/2008TC002267.
Sengupta, P., Arijit Ray, A., Pramanik, S., 2014. Mineralogical and chemical characteristics of newer dolerite dyke around Keonjhar, Orissa: Implication for hydrothermal activity in subduction zone setting. J. Earth Syst. Sci. 123, 887–904.
Shankar, R., Vijayagopal, B., Kumar, A., 2014. Precise Pb-Pb baddeleyite ages of 1765 Ma for a Singhbhum 'newer dolerite' dyke swarm. Curr. Sci. 106, 1306–1310.
Strik, G., Blake, T.S., Zegers, T.E., White, S.H., Langereis, C.G., 2003. Palaeomagnetism of flood basalt in the Pilbara Craton, Western Australia: late Archaean continental drift and the oldest known reversal of the geomagnetic field. J. Geophys. Res. 108. http://dx.doi.org/10.1029/2003JB002475.
Upadhyay, D., Sabyasachi Chattopadhyay, S., Kooijman, E., Mezger, K., Maity, J.B., 2014. Magmatic and metamorphic history of Paleoarcheantonalite–trondhjemite–granodiorite (TTG) suite fromthe Singhbhum craton, eastern India. Precambr. Res. 252, 180–190.
van Breemen, O., Davis, W.J., King, J.E., 1992. Temporal distribution of granitoid plutonic rocks in the Archean Slave Province, Northwest Canadian Shield. Can. J. Earth Sci. 29, 2186–2199.
Van Kranendonk, M.J., Hugh Smithies, R., Hickman, A.H., Champion, D.C., 2007. Review: secular tectonic evolution of Archean continental crust: interplay between horizontal and vertical processes in the formation of the Pilbara Craton, Australia. Terra Nova 19, 1–38.
Verma, R.K., Prasad, S.N., 1974. Paleomagnetic study and chemistry of newer Dolerites from Singhbhum, Bihar, India. Can. J. Earth Sci. 11, 1043–1054.
Walraven, F., Grobler, D.F., Key, R.M., 1996. Age equivalence of the Plantation Porphyry and the Kanye Volcanic Formation, south-eastern Botswana. S. Afr. J. Geol. 99, 23–31.
Wingate, M.T.D., 1999. Ion microprobe baddeleyite and zircon ages for Late Archaean mafic dykes of the Pilbara Craton, Western Australia. Aust. J. Earth Sci. 4, 493–500.
Wingate, M.T.D., 1998. A palaeomagnetic test of the Kaapvaal-Pilbara (Vaalbara) connection at 2.78 Ga. S. Afr. J. Geol. 4, 257–274.
Xie, H.Q., Hofmann, A., Hegner, E., Wilson, A., Wan, Y.S., Liu, D.Y., 2012. Zircon SHRIMP dating confirms a Palaeoarchaean supracrustal terrain in the southeastern Kaapvaal Craton, southern Africa. Gondwana Res. 21, 818–828.
Zegers, T.E., De Wit, M.J., Dann, J., White, S.H., 1998. Vaalbara, Earth's oldest assembled continent? A combined structural, geochronological, and palaeomagnetic test. Terra Nova 10, 250–259.